

Install SQLITE in Gitpod

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1. Open an empty gitpod workspace using <http://gitpod.new/>

New Workspace

Create a new workspace in the **bdur's Org** organization.



empty

github.com/gitpod-io/empty



VS Code · 1.97.2

Editor · Browser



Standard

Class · Up to 4 cores, 8GB RAM, 30GB storage



Continue (⌘Enter)



Starting



Opening Workspace ...

2. Install sqlite and sqlite web using

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

- **gitpod /workspace/empty** (main) \$ sudo apt install sqlite
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
sqlite is already the newest version (2.8.17-15fakesync1build1).
0 upgraded, 0 newly installed, 0 to remove and 1 not upgraded.
- **gitpod /workspace/empty** (main) \$ █

sudo apt install sqlite

pip install sqlite-web

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

- **gitpod /workspace/empty** (main) \$ ls
- **gitpod /workspace/empty** (main) \$ python
Python 3.12.9 (main, Feb 10 2025, 11:01:11) [GCC 11.4.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> exit()
- **gitpod /workspace/empty** (main) \$ pip install sqlite-web
Collecting sqlite-web
 Downloading sqlite-web-0.6.4.tar.gz (807 kB)
 807.6/807.6 kB 34.4 MB/s eta 0:00:00
 Preparing metadata (setup.py) ... done
Collecting flask (from sqlite-web)
 Downloading flask-3.1.0-py3-none-any.whl.metadata (2.7 kB)
Collecting peewee<=3.0.0 (from flask->sqlite-web)
 Downloading peewee-3.17.9.tar.gz (3.0 MB)
 3.0/3.0 MB 134.3 MB/s eta 0:00:00
 Installing build dependencies ... done
 Getting requirements to build wheel ... done
 Preparing metadata (pyproject.toml) ... done
Requirement already satisfied: pygments in /home/gitpod/.pyenv/versions/3.12.9/lib/python3.12/site-packages (from flask->sqlite-web) (2.19.1)
Collecting Werkzeug<=3.1 (from flask->sqlite-web)
 Downloading werkzeug-3.1.3-py3-none-any.whl.metadata (3.7 kB)
Requirement already satisfied: Jinja2<=3.1.2 in /home/gitpod/.pyenv/versions/3.12.9/lib/python3.12/site-packages (from flask->sqlite-web) (3.1.5)
Collecting itsdangerous<=2.2 (from flask->sqlite-web)
 Downloading itsdangerous-2.2.0-py3-none-any.whl.metadata (1.9 kB)
Collecting click<=8.1.3 (from flask->sqlite-web)
 Downloading click-8.1.8-py3-none-any.whl.metadata (2.3 kB)
Collecting blinker<=1.9 (from flask->sqlite-web)
 Downloading blinker-1.9.0-py3-none-any.whl.metadata (1.6 kB)
Requirement already satisfied: MarkupSafe<=2.0 in /home/gitpod/.pyenv/versions/3.12.9/lib/python3.12/site-packages (from Jinja2=>3.1.2->flask->sqlite-web) (3.0.2)
Downloading flask-3.1.0-py3-none-any.whl (102 kB)
Downloading blinker-1.9.0-py3-none-any.whl (8.5 kB)
Downloading click-8.1.8-py3-none-any.whl (98 kB)
Downloading itsdangerous-2.2.0-py3-none-any.whl (16 kB)
Downloading werkzeug-3.1.3-py3-none-any.whl (224 kB)
Building wheels for collected packages: sqlite-web, peewee
 Building wheel for sqlite-web (setup.py) ... done
 Created wheel for sqlite-web: filename=sqlite_web-0.6.4-py3-none-any.whl size=820316 sha256=0433f1442778cb115ff15b8af38dc16095ad9aacf61a791da821ffa95a36bb70
 Stored in directory: /workspace/.pyenv_mirror/pip_cache/wheels/58/18/ce/bb148273ed42beb91323c7ace3231b2e611f9e2aa7224a4f1b
 Building wheel for peewee (pyproject.toml) ... done
 Created wheel for peewee: filename=peewee-3.17.9-cp312-cp312-linux_x86_64.whl size=1021079 sha256=04c889e582d82a9c5c8ec8e2c9617b98617e6e0a00d3c47082cde10a8d2de895
 Stored in directory: /workspace/.pyenv_mirror/pip_cache/wheels/43/ef/2d/2c51d496bf084945ffdf838b4cc8767b8ba1cc20eb41588831
Successfully built sqlite-web peewee
Installing collected packages: peewee, Werkzeug, itsdangerous, click, blinker, flask, sqlite-web
Successfully installed Werkzeug-3.1.3 blinker-1.9.0 click-8.1.8 flask-3.1.0 itsdangerous-2.2.0 peewee-3.17.9 sqlite-web-0.6.4
- **gitpod /workspace/empty** (main) \$ █

3. Use command prompt of sqlite to create your first database with filename 'testsqlite.db'.

sqlite3 testsqlitebd.db

CREATE TABLE users (

id INTEGER PRIMARY KEY ,

name TEXT NOT NULL,

age INTEGER

);

insert into users values (1,"John",32);

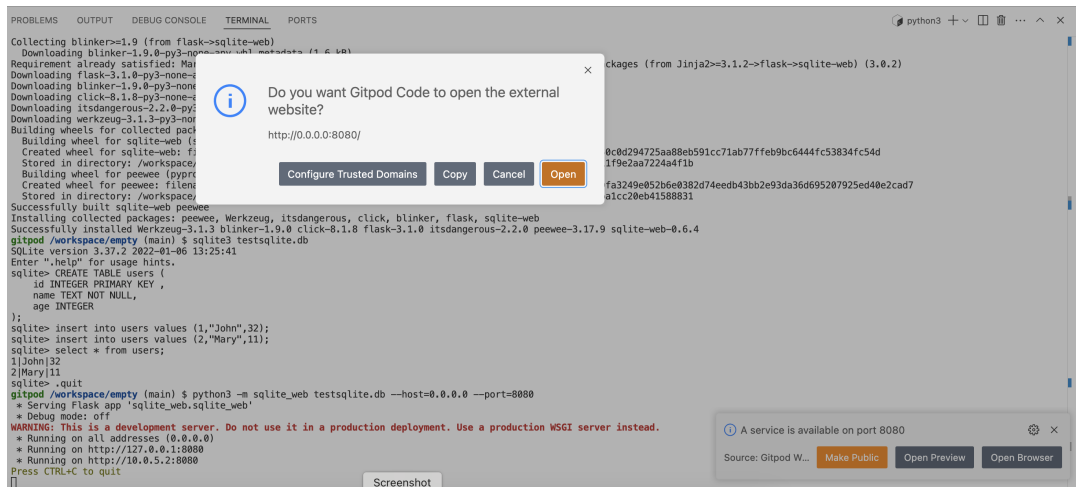
```
gitpod /workspace/empty (main) $ sqlite3 testsqlite.db
SQLite version 3.37.2 2022-01-06 13:25:41
Enter ".help" for usage hints.
sqlite> CREATE TABLE users (
  id INTEGER PRIMARY KEY ,
  name TEXT NOT NULL,
  age INTEGER
);
sqlite> insert into users values (1,"John",32);
sqlite> insert into users values (2,"Mary",11);
sqlite> select * from users;
1|John|32
2|Mary|11
sqlite> █
```

5. Lanuch sqlite-web with the 'testsqlite.db'. And show that you are able to visit the sqlite-web portal from your mobile phone.

IN the command terminal execute the following command:

python3 -m sqlite_web testsqlite.db --host=0.0.0.0 --port=8080

The symbol hyphen (-) is replace by underscore (_) in sqlite_web.



Port	Address	Protocol	Description	State
8080	https://8080-coral-wildfowl-3r33zvfhf5r.ws-us118.gitpod.io	HTTP		open (public)
36939	https://36939-coral-wildfowl-3r33zvfhf5r.ws-us118.gitpod.io	HTTP		open (private)

The status is made public, so the Web portal is accessible publicly.

testsqlite.db

new_table_name

Create

Query

table name...

users

[Toggle helper tables](#)

SQLite	3.37.2
Filename	/workspace/empty/testsqlite.db
Size	8.2 kB
Created	Friday February 28, 2025 at 09:58:PM
Modified	Friday February 28, 2025 at 09:58:PM
Tables	1
Indexes	0
Triggers	0
Views	0

Query

Execute SQL query...

Execute

6. Create a table and import the dataset ([flights-export.csv](#)) into testsqlite.db using sqlite-web.

flight_tab

Create

Query

Sqlite	3.37.2
Filename	/workspace/empty/testsqlite.db
Size	32.8 kB
Created	Sunday March 02, 2025 at 10:57:PM
Modified	Sunday March 02, 2025 at 09:02:PM
Tables	3
Indexes	0
Triggers	0
Views	0

Figure: create new table

ble

new_table_name

Create

Query

Structure

Content

Query

Insert

Drop

Import

Export

Import data

Input file

Choose File

testCSV_pandas.csv

Supported file types: **JSON** and **CSV**. JSON files must be structured as a list of objects. CSV files must contain a header row.

Create columns for unrecognized keys?

Yes

For instance, if your table has one column *name* and the import file contains objects with an additional column *age*, determine whether to store the *age* values as well.

Import

Cancel

Web-based SQLite database browser v0.6.4, powered by [Flask](#) and [Peewee](#). © 2025 [Charles Leifer](#)

Figure: upload csv file and create table

< > ↺

🔍 8080-coffee-weasel-ymzrfcyy01.ws-us118.gitpod.io/flight_tab/content/

🔗 🛡️ ⚠️

🔌 🔄 ⬇️ 📄 🔍 🌐 VPN

Update ⌵

testsqlite.db — flight_tab 66 rows, showing page 1

new_table_name

Create

Query

table name...

flight_tab

my_new_table

users

Toggle helper tables

Successfully imported 66 objects from flights-export-1.csv.

Structure

Content

Query

Insert

Drop

Import

Export

id	Num_of_Passengers	Quarter	Weather_Natural_Disaster	Q_Y	Avg_Airfare	Avg_Comp_Airfare	Q3	Price_Alt_Transport_Gasoline_	Entry_Exit_in_Market	Recession	GDP	COVID_19_	Q2	N
1	554	1	0	Q1 2006	232.6	170	0	2.34	3	0	13599.16	0	0	-
2	1283	2	0	Q2 2006	206.67	164.4	0	2.84	4	0	13753.424	0	1	1
3	727	3	0	Q3 2006	276.86	200.25	1	2.83	3	0	13870.188	0	0	-
4	677	4	0	Q4 2006	183	214.8	0	2.26	3	0	14039.56	0	0	-
5	579	1	0	Q1 2007	230	216.25	0	2.36	3	0	14215.651	0	0	-
6	453	2	0	Q2 2007	250.4	185.67	0	3.02	3	0	14402.082	0	1	8

Figure: Few rows of the table

testsqlite.db — flight_tab

table name...	Structure	Content	Query	Insert
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flight_tab

my_new_table

users

Toggle helper tables

SQL

```
CREATE TABLE "flight_tab" (  
  "id" INTEGER NOT NULL PRIMARY KEY,  
  "Q2" TEXT,  
  "GDP" TEXT,  
  "Price_Alt_Transport_Gasoline_" TEXT,  
  "Q3" TEXT,  
  "Net_Income_Fin_Position_" TEXT,  
  "Entry_Exit_in_Market" TEXT,  
  "Num_of_Passengers" TEXT,  
  "Avg_Airfare" TEXT,  
  "Weather_Natural_Disaster" TEXT,  
  "Year" TEXT,  
  "Recession" TEXT,  
  "Q_Y" TEXT,  
  "Q1" TEXT,  
  "Quarter" TEXT,  
  "Avg_Comp_Airfare" TEXT,  
  "COVID_19_" TEXT  
)
```

Figure: structure of the table


```
CREATE TABLE "flight_tab" (  
  "id" INTEGER NOT NULL PRIMARY KEY,  
  "Q2" TEXT,  
  "GDP" TEXT,  
  "Price_Alt_Transport_Gasoline_" TEXT,  
  "Q3" TEXT,  
  "Net_Income_Fin_Position_" TEXT,  
  "Entry_Exit_in_Market" TEXT,  
  "Num_of_Passengers" TEXT,  
  "Avg_Airfare" TEXT,  
  "Weather_Natural_Disaster" TEXT,  
  "Year" TEXT,  
  "Recession" TEXT,  
  "Q_Y" TEXT,  
  "Q1" TEXT,  
  "Quarter" TEXT,  
  "Avg_Comp_Airfare" TEXT,  
  "COVID_19_" TEXT  
)
```

7. In sqlite-web use SQL statement to finish the following queries.

a) Calculate the total and average number of passengers per year, given that Net Income (Fin. Position) > 0.

SELECT Year, SUM(CAST(Num_of_Passengers AS INTEGER)) AS total_passengers,
AVG(CAST(Num_of_Passengers AS INTEGER)) AS average_passengers
FROM flight_tab

WHERE
CAST(Net_Income_Fin_Position_ AS
INTEGER) >
0 GROUP
BY Year;

```
SELECT Year, SUM(CAST(Num_of_Passengers AS INTEGER)) AS total_passengers,  
AVG(CAST(Num_of_Passengers AS INTEGER)) AS average_passengers  
FROM flight_tab  
WHERE CAST(Net_Income_Fin_Position_ AS INTEGER) > 0 GROUP BY Year;
```

Use Shift + Up/Down to navigate recently-executed queries

Execute

Export JSON

Export CSV

SQL Help ▾

Boo

Results (unable to determine)

Year	total_passengers	average_passengers
2006	1283	1283.0
2007	453	453.0
2010	1045	348.33333333333333
2011	396	396.0
2012	434	434.0
2013	943	471.5
2014	7019	1754.75
2015	97199	24299.75
2016	160312	40078.0
2017	159085	39771.25
2018	174378	43594.5
2019	188020	47005.0
2021	63425	31712.5
2022	51953	51953.0

b) Calculate the total and average number of passengers per year, given that Net Income (Fin. Position) < 0.

```
SELECT Year, SUM(CAST(Num_of_Passengers AS INTEGER)) AS total_passengers,
        AVG(CAST(Num_of_Passengers AS INTEGER)) AS average_passengers
FROM flight_tab
WHERE CAST(Net_Income_Fin_Position_ AS INTEGER) < 0
GROUP BY Year;
)
```

```
SELECT Year, SUM(CAST(Num_of_Passengers AS INTEGER)) AS total_passengers,
        AVG(CAST(Num_of_Passengers AS INTEGER)) AS average_passengers
FROM flight_tab
WHERE CAST(Net_Income_Fin_Position_ AS INTEGER) < 0
GROUP BY Year;
```

Use Shift + Up/Down to navigate recently-executed queries

Execute

Export JSON

Export CSV

SQL Help ▾

Boc

Results (unable to determine)

Year	total_passengers	average_passengers
2006	1958	652.6666666666666
2007	1389	463.0
2008	1820	455.0
2009	1422	355.5
2010	472	472.0
2011	1468	489.3333333333333
2012	2234	744.6666666666666
2013	1257	628.5
2020	60574	15143.5
2021	50567	25283.5
2022	39742	39742.0

c) Calculate the total and average number of passengers per year.

SELECT

Year, SUM(CAST(Num_of_Passengers AS INTEGER)) AS total_passengers,

AVG(CAST(Num_of_Passengers AS INTEGER)) AS average_passengers

FROM flight_tab GROUP BY Year;

```
SELECT
  Year, SUM(CAST(Num_of_Passengers AS INTEGER)) AS total_passengers,
  AVG(CAST(Num_of_Passengers AS INTEGER)) AS average_passengers
FROM flight_tab GROUP BY Year;
```

Use Shift + Up/Down to navigate recently-executed queries

Execute

Export JSON

Export CSV

SQL Help ▾

Bool ▾

Results (unable to determine)

Year	total_passengers	average_passengers
2006	3241	810.25
2007	1842	460.5
2008	1820	455.0
2009	1422	355.5
2010	1517	379.25
2011	1864	466.0
2012	2668	667.0
2013	2200	550.0
2014	7019	1754.75
2015	97199	24299.75
2016	160312	40078.0
2017	159085	39771.25
2018	174378	43594.5
2019	188020	47005.0
2020	60574	15143.5
2021	113992	28498.0
2022	91695	45847.5

8. Use sqlite command prompt to redo one of the queries in 7.

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
gitpod /workspace/empty (main) $ sqlite3 testsqlite.db
SQLite version 3.37.2 2022-01-06 13:25:41
Enter ".help" for usage hints.
sqlite> .tables
flight_tab    my_new_table    users
sqlite> SELECT
    Year, SUM(CAST(Num_of_Passengers AS INTEGER)) AS total_passengers,
    AVG(CAST(Num_of_Passengers AS INTEGER)) AS average_passengers
FROM flight_tab    GROUP BY Year;
2006|3241|810.25
2007|1842|460.5
2008|1820|455.0
2009|1422|355.5
2010|1517|379.25
2011|1864|466.0
2012|2668|667.0
2013|2200|550.0
2014|7019|1754.75
2015|97199|24299.75
2016|160312|40078.0
2017|159085|39771.25
2018|174378|43594.5
2019|188020|47005.0
2020|60574|15143.5
2021|113992|28498.0
2022|91695|45847.5
sqlite> █
```